Proof

Chapter 7 · Original slide 24

4: Proof

24

Terminology

A **conjecture** is a mathematical statement that has yet to be proven.

One famous conjecture is **Goldbach's Conjecture**.
It states "*Every even integer greater than 2 can be expressed as the sum of two primes.*"
It has been verified up to 4×10^{18} (that's big!); this provides evidence that it is true, but does not prove it is true!

A **theorem** is a mathematical statement that has been proven.

One famous misnomer was **Fermat's Last Theorem**, which states "*If n is an integer where $n > 2$, then $a^n + b^n = c^n$ has no non-zero integer solutions for a, b, c.*" It was 300 years until this was proven in 1995. Only then was the 'Theorem' in the name then correct!

Chapter 7 · Original slide 25

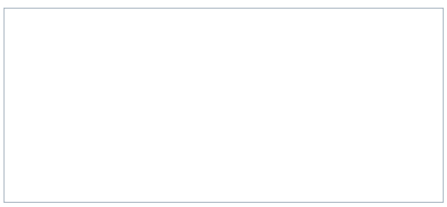
Types of Proof

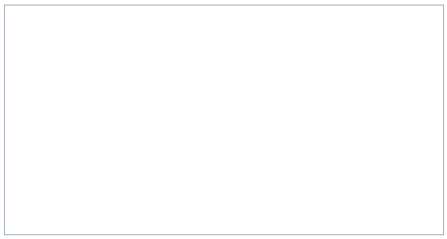
25

A proof must show all **assumptions** you are using, have a clear **sequential list of steps** that logically follow, and must cover **all possible cases**.
You should usually make a **concluding statement**, e.g. restating the original conjecture that you have proven.

a. Proof by Deduction

This is the simplest type, where you start from known facts and reach the desired conclusion via deductive steps.





Proof

Chapter 7 · Original slide 26

Be Warned…

26

Proof by Deduction requires you to **start from known facts** and end up at the conclusion. It is **not** acceptable to start with to the conclusion, and verify it works, **because you are assuming the thing you are trying to prove**.

Example: Prove that if three consecutive integers are the sides of a right-angled triangle, they must be 3, 4 and 5.

Incorrect Proof:

Correct Proof:

Chapter 7 · Original slide 27

Types of Proof

27

a. Proof by Deduction

Exam Tip: This is quite a common last parter.

Prove that x^2+4x+5 is positive for all values of x.

Test Your Understanding

Prove that the sum of the squares of two consecutive odd numbers is 2 more than a multiple of 8.

$2n-1$ $2n+1$

Proof

Chapter 7 · Original slide 28

Exercise 7D – Y1 Pages 149-150

28

11 Prove that $x^2 + 8x + 20 \geq 4$ for all values of x.

12 The equation $kx^2 + 5kx + 3 = 0$, where k is a constant, has no real roots. Prove that k satisfies the inequality $0 \leq k < \frac{12}{25}$. \hfill(4 mar ...

Chapter 7 · Original slide 29

Exercise 7D – Y1 Pages 149-150

29

13 The equation $px^2 - 5x - 6 = 0$, where p is a constant, has two distinct real roots. Prove that p satisfies the inequality $p > -\frac{25}{24}$. \hfill(4 mar ...

18 A circle has equation $(x-1)^2 + y^2 = k$, where $k > 0$.
The straight line L with equation $y = ax$ cuts the circle at two distinct points.
Prove that $k > \frac{a^2}{1+a^2}$.

Proof

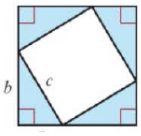
Chapter 7 · Original slide 30

Exercise 7D – Y1 Pages 149-150

30

19 Prove that the line $4y - 3x + 26 = 0$ is a tangent to the circle $(x + 4)^2 + (y - 3)^2 = 100$.

20 The diagram shows a square and four congruent right-angled triangles.
Use the diagram to prove that $a^2 + b^2 = c^2$.



Chapter 7 · Original slide 32

Other Types of Proof

32

b. Proof by Exhaustion

This means breaking down the statement into **all possible smaller cases**, where we prove each individual case.
(This technique is sometimes known as 'case analysis')

Prove that $\mathbf{n^2+n}$ is even for all integers $\mathbf{n}$.

c. Disproof by Counter-Example

While to prove a statement is true, we need to prove every possible case (potentially infinitely many!), **we only need one example to disprove** a statement.
This is known as a **counterexample**.

Disprove the statement:
"$\mathbf{n^2-n+41}$ is prime for all integers $\mathbf{n}$."

Proof

Chapter 7 · Original slide 33

Exercise 7E – Y1 Pages 152-154

33

4 Prove that all cube numbers are either a multiple of 9 or 1 more or 1 less than a multiple of 9.

7 Prove that for all real values of x
$(x+6)^2 \geq 2x+11$

Chapter 7 · Original slide 34

Exercise 7E – Y1 Pages 152-154

34

8 Given that a is a positive real number, prove that:
$a + \frac{1}{a} \geq 2$

9 a Prove that for any positive numbers p and q:
$p+q > \sqrt{4pq}$

b Show, by means of a counter-example, that this inequality does not hold when p and q are both negative. \hfill(2 m . . .